DEBT SUSTAINABILITY I

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PLAN

- We review several approaches to debt sustainability analysis.
- Can the government service its current outstanding debt? Or will a fiscal consolidation be needed?
- Is the outstanding public debt and its projected path consistent with those of the government's revenues and expenditures?

CLASSIC DEBT SUSTAINABILITY ANALYSIS

GOVERNMENT BUDGET CONSTRAINT

- The government flow budget constraint:

$$B_{t+1} = R_{t+1} B_t + G_{t+1} - T_{t+1},$$

  where B_t is beginning-of-period nominal debt, $R_{t+1} \equiv 1 + i_{t+1}$ is the gross nominal return on government debt, and $G_{t+1} - T_{t+1}$ is the nominal primary deficit.
- The budget constraint is written in nominal terms. The debt-to-GDP ratio $b_t = B_t / Y_t$ is unit-free (inflation cancels in the ratio).
- Denote the primary deficit $PD_{t+1} \equiv G_{t+1} - T_{t+1}$ and the primary surplus $S_{t+1} \equiv T_{t+1} - G_{t+1} = -PD_{t+1}$.
- We abstract from money growth (we can treat money as a type of government debt).
- Note: is it really a “constraint”?

DEBT TO OUTPUT RATIO

- Define the gross nominal growth rate $\Gamma_{t+1} \equiv Y_{t+1}/Y_t$. Divide by Y_{t+1}:

$$b_{t+1} = \frac{R_{t+1}}{\Gamma_{t+1}} b_t + pd_{t+1},$$

  where $b_t \equiv B_t/Y_t$ and $pd_t \equiv (G_t - T_t)/Y_t$.

- Define cumulative objects:

$$\mathcal{R}_{s:t} \equiv \prod_{k=s+1}^{t} R_k, \qquad \mathcal{G}_{s:t} \equiv \prod_{k=s+1}^{t} \Gamma_k = \frac{Y_t}{Y_s}.$$

DEBT TO OUTPUT RATIO

- For simplicity assume $B_0 = 0$.
- Iterating backward:

$$b_{t+1} = \sum_{s=1}^{t+1} \frac{\mathcal{R}_{s:t+1}}{\mathcal{G}_{s:t+1}} pd_s.$$

- Debt to output ratio today is determined by:
  1. The past primary deficits to GDP ratios;
  2. The past returns on debt;
  3. The past growth rates of (nominal) GDP.
- We can use the formula above to understand what determines the current debt to output ratio.

CLASSIC DEBT SUSTAINABILITY ANALYSIS

- A version of the formula is often used to assess debt sustainability – "whether the government can service its debt".
- Warning: this is about the future, not the present. The fact that people use the formula to assess the current situation is often a red flag!
- Classic debt sustainability analysis looks at the "long run".

CLASSIC DEBT SUSTAINABILITY ANALYSIS

- Assume that the economy is in a steady state with constant $\bar{\Gamma}$, $\bar{R}$, and a constant primary deficit ratio $\overline{pd}$.
- What is the debt to output ratio consistent with the above?
- If $\bar{\Gamma} > \bar{R}$: the steady state is stable, and debt converges to b^{SS} from any starting point.
- If $\bar{R} > \bar{\Gamma}$: the steady state is unstable – any deviation from b^{SS} leads to explosive dynamics. Sustainability requires the surplus to be at least s^*.

CLASSIC DEBT SUSTAINABILITY ANALYSIS

- Classic debt sustainability analysis usually analyzes deterministic setups (or perfect foresight) with one-period debt.
- In these setups, R_{t+1} is simply $1 + i^f$, the gross nominal risk-free rate.
- Setting $R_{t+1} = \bar{R}$ and $\Gamma_{t+1} = \bar{\Gamma}$ constant, the steady state of $b_{t+1} = (\bar{R}/\bar{\Gamma})\, b_t + \overline{pd}$ is

$$b^{ss} = \overline{pd}\, \frac{\bar{\Gamma}}{\bar{\Gamma} - \bar{R}}.$$

- This requires $\bar{\Gamma} \neq \bar{R}$. The sign of $\bar{\Gamma} - \bar{R}$ determines whether the government needs surpluses or can run deficits.

CLASSIC DSA: MAASTRICHT

- The Treaty of Maastricht set the limit of the debt to output ratio at 60% and the total deficit to output ratio at 3%.
- Caution: the Maastricht deficit includes interest payments. Define the total deficit ratio

$$d_{t+1} \equiv \frac{G_{t+1} + i_{t+1} B_t - T_{t+1}}{Y_{t+1}} = \frac{i_{t+1} b_t}{\Gamma_{t+1}} + p d_{t+1}.$$

- Since $B_{t+1} = B_t + D_{t+1}$, dividing by Y_{t+1} gives

$$b_{t+1} = \frac{b_t}{\Gamma_{t+1}} + d_{t+1}.$$

In steady state ($x \equiv \bar{\Gamma} - 1$): $b = \bar{d} \frac{\bar{\Gamma}}{\bar{\Gamma}-1} \approx \frac{\bar{d}}{x}$.

CLASSIC DSA: MAASTRICHT

$$b \approx \frac{\bar{d}}{x}$$

- For $b = 0.6$ and $\bar{d} = 0.03$: $x \approx 0.03/0.6 = 0.05$, i.e. 5% nominal GDP growth.
- For the EU in the 1990s this meant roughly 3% real growth plus 2% inflation – plausible, but increasingly unrealistic for many member states.
- Note: this result is independent of $\bar{R}$ – the interest cost is already embedded in $\bar{d}$.
- If the actual growth rate is lower, the debt to output ratio will be higher, even if the deficit stays at the limit.

CLASSIC DSA: STABILIZING THE DEBT RATIO

- What primary surplus $s \equiv -pd$ is needed to keep b constant?
- Setting $b_{t+1} = b_t = b$ in $b_{t+1} = (\bar{R}/\bar{\Gamma})\, b - s$:

$$s^* = \frac{\bar{R} - \bar{\Gamma}}{\bar{\Gamma}}\, b.$$

  - $\bar{R} > \bar{\Gamma}$: must run a surplus proportional to debt – the "debt burden".
  - $\bar{R} < \bar{\Gamma}$: can run a deficit and still keep debt stable – growth does the work.
  - The required adjustment is larger the higher is b: high debt requires large surpluses.
- This formula is the workhorse of IMF and ECB debt sustainability assessments.

CLASSIC DSA: DEBT DYNAMICS

- Suppose the government runs a constant surplus s every period. Starting from b_0:

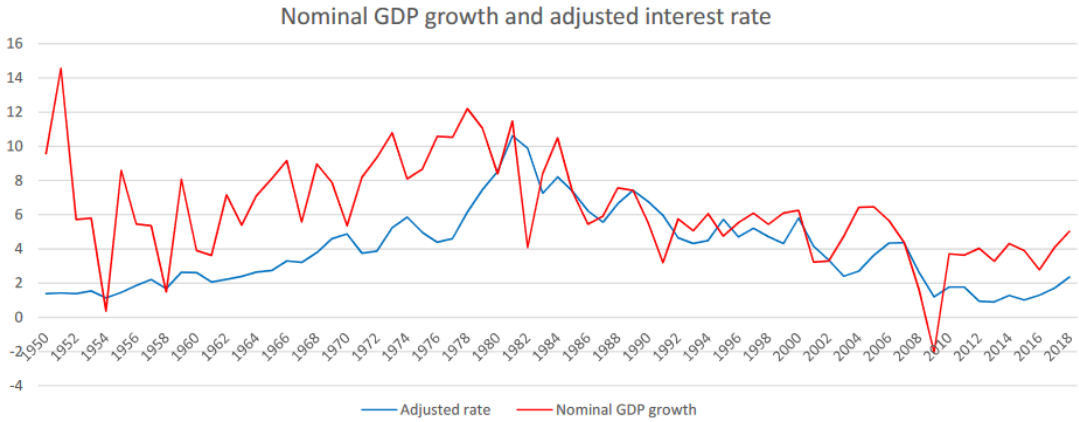
$$b_{t+1} = \left(\frac{\bar{R}}{\bar{\Gamma}}\right)^{t+1}\left(b_0 - b^{ss}\right) + b^{ss}, \quad b^{ss} = \overline{pd}\,\frac{\bar{\Gamma}}{\bar{\Gamma} - \bar{R}}.$$

- Two cases depending on $\bar{R}$ versus $\bar{\Gamma}$:
  1. $\bar{R} > \bar{\Gamma}$: $(\bar{R}/\bar{\Gamma})^t \to \infty$ – debt explodes unless $s \geq s^*$. This is the snowball effect: interest accrues faster than the economy grows.
  2. $\bar{R} < \bar{\Gamma}$: $(\bar{R}/\bar{\Gamma})^t \to 0$ – debt converges to b^{ss} regardless of b_0. Any constant surplus (even a small deficit) stabilizes debt.
- Key takeaway: $\bar{R} \gtrless \bar{\Gamma}$ determines whether debt dynamics are explosive or self-correcting.

CLASSIC DEBT SUSTAINABILITY ANALYSIS

- If $\bar{\Gamma} > \bar{R}$ it seems there is no fiscal cost to debt.
- Note: $\bar{R}/\bar{\Gamma} = (1+r)/(1+g)$ where r and g are real – inflation cancels. So $\bar{\Gamma} > \bar{R} \Leftrightarrow g > r$: the "$r$ versus g" debate.
- Blanchard (2019) argues that $\bar{\Gamma} > \bar{R}$ is a norm, not an exception.
- But what is $\bar{R}$? How to measure it? Blanchard (2019) looks at the 1-year US Treasury bill rate, the 10-year US Treasury bond rate, adjusts for various maturities…
- Note: it does not necessarily mean that it is optimal to have deficits.

RATES IN THE US



Source: Blanchard (2019)

WHY DOES $r < g$ MATTER? DIAMOND (1965)

- If $r < g$ (equivalently $\bar{R} < \bar{\Gamma}$), the economy may be dynamically inefficient (Diamond 1965).
- Dynamic inefficiency: capital stock is too large; the economy overaccumulates. Reducing investment raises consumption for *every* generation.
- Government debt crowds out capital – but here crowding out is welfare-improving: debt absorbs excess savings and moves the economy toward the Golden Rule.

BLANCHARD (2019): WELFARE EFFECTS OF DEBT ARE AMBIGUOUS

- Fiscal cost of debt is zero when $\bar{R} < \bar{\Gamma}$: debt rollovers are feasible.
- But what about welfare? Two channels work in opposite directions:
  1. Safe rate channel: if the real safe rate $< g$, an intergenerational transfer via debt raises welfare.
  2. Risky rate channel: if the real marginal product of capital $> g$ (which it is empirically), crowding out capital lowers welfare.
- The net effect is ambiguous – welfare cost may be smaller than commonly thought, but it is not necessarily negative.

IS THE US DYNAMICALLY INEFFICIENT? ABEL ET AL. (1989)

- The real safe rate being below g does not imply dynamic inefficiency.
- Abel et al. (1989): the relevant test compares gross capital income (profits, rents, interest) to gross investment.
- If capital income exceeds investment, the economy is dynamically efficient – capital is not overaccumulated.
- The US passes this test: the economy is dynamically efficient despite the real safe rate being below g.
- Key distinction: the real safe rate can be below g while the real marginal product of capital is above g.

A GOLDILOCKS THEORY OF FISCAL DEFICITS (MIAN, STRAUB, SUFI 2022)

- Blanchard (2019): $\bar{R} < \bar{\Gamma} \Rightarrow$ deficit is free. Not so fast.
- Mian, Straub, Sufi (2022): $\bar{R} < \bar{\Gamma}$ arises from a convenience yield on government bonds – investors value their safety and liquidity, so they accept a return below the growth rate.
- Inequality amplifies this: when the rich save more (high wealth concentration), the convenience yield rises, pushing R further below Γ.
- But as the government issues more debt, the convenience yield shrinks (bonds become less scarce), pushing R back toward Γ. This tightens the government's own budget constraint.
- There is a Goldilocks zone: a moderate deficit that
  1. exploits the convenience yield (keeps $R < \Gamma$);
  2. does not push debt so high that the convenience yield vanishes and R catches up with Γ.

A GOLDILOCKS THEORY OF FISCAL DEFICITS (CONT.)

- Key mechanism: the deficit–debt locus is hump-shaped. Small permanent deficits are sustainable; large ones cause debt to explode because R rises endogenously toward Γ.
- Implication: even when $\bar{R} < \bar{\Gamma}$, there is an optimal, finite deficit. The free-lunch argument breaks down once you model *why* $r < g$ holds.
- Contrast with Diamond (1965): Diamond's welfare gain from debt relies on crowding out *excess capital*. MSS show the same $\bar{R} < \bar{\Gamma}$ can arise from convenience yields with no overaccumulation of capital – the welfare arithmetic is different.
- Related: "Indebted Demand" (MSS, QJE 2021) studies how inequality drives down r through excess saving *and* generates fragile demand sustained by household debt. The two papers share the theme but use different models.

CLASSIC DSA: WHAT IT CANNOT TELL US

- It only defines what long-run debt is for a given long-run primary balance (or vice versa) if stationarity holds, or defines lower bounds on the short-run dynamics of the primary balance.
- It does not pin down which steady state b^{SS} the economy should converge to – there is no optimality criterion built in.
- For a given initial b_{t-1}, there are multiple paths of the primary balance that satisfy the IGBC – with different values of b^{SS} and different transitional dynamics.
- Some of these paths converge to a finite steady state; others let debt diverge to infinity – yet all satisfy the IGBC (which we derive next)!

CLASSIC DSA: WHAT IT CANNOT TELL US (CONT.)

- IGBC: the value of debt is equal to the present discounted value of future primary surpluses.
- Wrong discount rate: discounting at the risk-free rate $\bar{R}$ is appropriate only under perfect foresight or risk-neutrality. If surpluses are procyclical (fall in recessions), investors demand a risk premium – discounting at $\bar{R}$ overstates the present value of surpluses and understates the fiscal burden.
- Cannot rank paths: even when multiple fiscal adjustment paths all satisfy the IGBC, they have very different implications for output, prices, and welfare. The classic non-structural approach cannot evaluate which path is better – it only checks whether the constraint is met, not what the adjustment costs.

INTERTEMPORAL GOVERNMENT BUDGET CONSTRAINT

IGBC

- We used the government budget constraint by solving it backward (go back in time).
- We can also solve it forward – the valuation approach, the market value of government debt is determined by the discounted value of future government surpluses.
- This idea is often used in finance (e.g., Campbell and Shiller 1988) - what determines stock prices?
- Allows us to think seriously about risk and asset pricing.

IGBC

- Ignore risk for now and rewrite $B_{t+1} = R_{t+1} B_t + PD_{t+1}$ as

$$B_t = \frac{1}{R_{t+1}} \left(B_{t+1} + S_{t+1} \right).$$

- Continue this process to get

$$B_t = \sum_{j=1}^{T} \frac{S_{t+j}}{\mathcal{R}_{t:t+j}} + \frac{B_{t+T}}{\mathcal{R}_{t:t+T}}.$$

IGBC

- Note: there is a term $B_{t+T}/\mathcal{R}_{t:t+T}$.
- If this term does not vanish as $T \to \infty$, there is a bubble.
- Imposing the no-Ponzi game condition ($\lim_{T\to\infty} B_{t+T}/\mathcal{R}_{t:t+T} = 0$) we get the IGBC:

$$B_t = \sum_{j=1}^{\infty} \frac{S_{t+j}}{\mathcal{R}_{t:t+j}}.$$

- If the above is satisfied, we say that IGBC holds.

NO-PONZI CONDITION: WHAT ENFORCES IT?

- The no-Ponzi condition $\lim_{T\to\infty} B_{t+T}/\mathcal{R}_{t:t+T} = 0$ is imposed, not derived. What makes it hold?
- For households: lenders refuse to roll over debt forever – the transversality condition from the household optimisation problem.
- For governments: no analogous private enforcement mechanism. Governments can (and do) default or inflate away debt.

NO-PONZI CONDITION: THREE INTERPRETATIONS

- Three possible interpretations of what enforces no-Ponzi for governments:
  1. Market discipline: investors stop buying debt if they expect Ponzi finance – self-fulfilling rollover crises (Calvo 1988).
  2. Political constraint: constitutional or treaty limits (e.g. Maastricht) substitute for market enforcement.
  3. Fiscal theory: the price level adjusts so that the real value of debt equals the PDV of surpluses – the constraint always holds *in real terms*.
- Bottom line: the no-Ponzi condition is an equilibrium outcome, not a physical constraint.

WHEN THE IGBC FAILS: DEFAULT

- If no feasible sequence of surpluses can satisfy the IGBC at current prices, the government must default (explicitly or via inflation).
- Fiscal fatigue (Ghosh et al. 2013): empirically, the FRF response ρ falls at high debt levels. There exists a debt limit $\bar{b}$ above which no politically feasible surplus can stabilize debt.
- The gap between observed b and the debt limit $\bar{b}$ is sometimes called fiscal space – the room for fiscal manoeuvre before default risk becomes binding.

WHY DEBT CRISES ARE SUDDEN

- Feedback loop: as $b \to \bar{b}$, investors demand higher spreads $\Rightarrow$ effective R rises $\Rightarrow$ the required surplus s^* increases $\Rightarrow$ sustainability worsens further.
- This is a vicious circle: doubt about solvency raises borrowing costs, which makes solvency harder to achieve, which raises doubt further.
- A country can look solvent one day and face a crisis the next – the feedback loop amplifies small shocks into large ones.

NOMINAL VS. REAL SOLVENCY

- “The government can always print money to repay debt.”
- Nominally correct: a sovereign issuing debt in its own currency need not default in nominal terms. This is the kernel of truth in MMT.
- But monetisation generates seigniorage — an inflation tax on money holders — and the revenue is bounded by the money-demand Laffer curve (Cagan 1956).

NOMINAL VS. REAL SOLVENCY (CONT.)

- Three adjustment margins, all with finite capacity:
  1. Primary surpluses (bounded by fiscal fatigue)
  2. Seigniorage (bounded by the Laffer curve)
  3. Explicit default (haircut on creditors)
- The IGBC in real terms still binds $\Rightarrow$ the debt limit $\bar{b}$ is finite even for a sovereign currency issuer.
- Printing money changes *who pays*, not *whether* there is a constraint.

IGBC WITH RISK

- Under uncertainty the IGBC becomes

$$B_t = \mathbb{E}_t \sum_{j=1}^{\infty} M_{t,t+j} S_{t+j},$$

  where $M_{t,t+j}$ is the nominal stochastic discount factor (SDF) – it prices nominal payoffs – and $S_{t+j} = T_{t+j} - G_{t+j}$ is the nominal primary surplus.
- $M_{t,t+j}$ reflects how holders of government debt discount future cash flows.
- It is a function of the state of the economy at t and $t + j$. Recall the first-order condition from the household problem.

RISK AND THE DISCOUNT RATE

- Define the nominal risk-free cumulative discount factor $\mathcal{R}^f_{t:t+j} \equiv 1/\mathbb{E}_t[M_{t,t+j}]$.
- By the covariance decomposition:

$$\mathbb{E}_t[M_{t,t+j}\, S_{t+j}] = \frac{\mathbb{E}_t\ S_{t+j}}{\mathcal{R}^f_{t:t+j}} + \mathrm{Cov}_t\big(M_{t,t+j},\, S_{t+j}\big)\,.$$

- Hence discounting surpluses at the risk-free rate is valid only if $\mathrm{Cov}_t(M, S) = 0$.

RISK AND THE DISCOUNT RATE (CONT.)

- If surpluses are procyclical (S_{t+j} falls in bad times when $M_{t,t+j}$ is high): $\text{Cov}_t(M, S) < 0 \Rightarrow$ discounting at $\mathcal{R}^f$ overstates the PDV of surpluses.
- Implication: Classic DSA and the deterministic IGBC are not reliable guides to true fiscal space.
- Next lecture: how to measure the correct SDF and risk-adjusted fiscal space (asset-pricing approach to debt sustainability).

DEBT SUSTAINABILITY

- We can say that debt is sustainable if and only if the IGBC holds.
- Problem: this condition is about the entire future.
- Solution (?): use forecasts of future s_t to compute the present value of future surpluses. Some early papers did this, but they used risk-free rates.
- Valid if one of these conditions holds:
  1. There is perfect foresight;
  2. Investors are risk-neutral;
  3. Primary surpluses do not covary with the SDF.

BOHN (1998)

- Not even $b_t \to \infty$ means that the IGBC does not hold – debt has to go to infinity slowly enough.
- Bohn (1998): see if the government does something that guarantees the IGBC holds, investigate the fiscal reaction function.
- Allows us to sidestep the problem of forecasting future surpluses and choosing the correct discount rate.
- Sufficient condition: the IGBC might hold even if Bohn's condition is not satisfied, but if the condition *is* satisfied, the IGBC holds for sure.

BOHN (1998)

- Linear fiscal reaction function:

$$s_t = \rho\, b_{t-1} + z_t + \epsilon_t.$$

- s_t is the primary surplus ratio; z_t captures other determinants of the surplus.
- Check if $\rho > 0$ – the government raises the surplus when debt is high.
- If $\rho > 0$, then the IGBC holds even if ρ is below $\bar{R}/\bar{\Gamma} - 1$.

BOHN (1998)

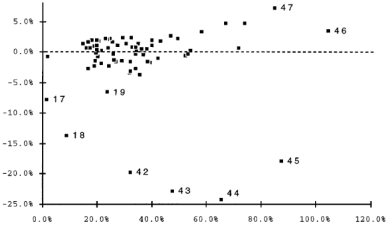
- If $\rho > 0$, the IGBC holds for any initial level of debt.
- This analysis works also for $\bar{R} = \bar{\Gamma}$ – there was division by zero in the classic analysis.
- If $\bar{R}/\bar{\Gamma} - 1 > \rho > 0$, debt explodes, but the IGBC still holds (under certain conditions: see Bohn 2007).

BOHN (1998)

- Bohn (1998) estimates ρ for the US in 1916–1995.
- He includes the level of temporary government spending and a business cycle indicator in z_t.
- He finds a positive value of ρ, around 0.05 for the entire sample.

BOHN (1998)

(a) The simple correlation



(b) With adjustment for temporary spending and output fluctuations

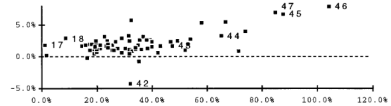


FIGURE I

Primary Surplus versus Initial Debt

The graph shows the privately held government debt/GDP at the start of a period on the horizontal axis against the primary budget surplus/GDP on the vertical axis, for 1916–1995; (a) shows raw data, and (b) shows the adjusted surplus, as explained in the text.

Source: Bohn (1998)

BOHN (1998)

TABLE I
DETERMINANTS OF THE BUDGET SURPLUS

Dependent variable primary budget surplus divided by GDP (s_t)							
Sample	Constant	*G*VAR	*Y*VAR	d_t	R^2	σ	*DW*
(1) 1916–1995	−0.019	−0.776	−1.450	0.054	0.936	0.014	1.42
	(−5.424)	(−33.001)	(−3.628)	(6.048)			
	[−3.957]	[−20.874]	[−4.075]	[3.787]			
(2) 1920–1995 excl. 1940–1947	−0.009	−0.551	−1.906	0.028	0.618	0.011	1.40
	(−2.030)	(−4.034)	(−4.666)	(2.701)			
	[−2.155]	[−3.721]	[−4.296]	[2.491]			
(3) 1916–1983	−0.018	−0.782	−1.414	0.054	0.942	0.014	1.54
	(−4.903)	(−31.667)	(−3.360)	(5.996)			
	[−3.958]	[−20.943]	[−4.004]	[4.076]			
(4) 1920–1982 excl. 1940–1947	−0.008	−0.520	−1.912	0.030	0.630	0.011	1.56
	(−1.710)	(−3.612)	(−4.441)	(2.815)			
	[−1.932]	[−3.272]	[−3.959]	[2.856]			
(5) 1948–1995	−0.015	−0.593	−2.139	0.037	0.651	0.010	1.54
	(−3.536)	(−4.182)	(−4.361)	(3.589)			
	[−3.496]	[−3.701]	[−3.757]	[2.821]			
(6) 1960–1984	−0.013	−0.410	−2.051	0.044	0.724	0.007	1.43
	(−2.110)	(−2.173)	(−4.174)	(2.028)			
	[−2.174]	[−2.281]	[−3.391]	[2.587]			

The variable d_t is the privately held debt/GDP at the start of the year. *G*VAR and *Y*VAR are measures of temporary government spending and of cyclical variations in output, respectively, from Barro [1986a]. All estimates are OLS with annual data; ( ) = ordinary *t*-statistics; [ ] = heteroskedasticity- and autocorrelation-consistent *t*-statistics (computed with Newey-West lag window of size 1); σ = standard error; *DW* = Durbin-Watson statistic.

Source: Bohn (1998)

FISCAL REACTION FUNCTIONS

- Bohn (2008) extends the analysis to 1792–2003. He finds $\rho > 0.1$, more than twice as large as in the earlier study.
- Mendoza and Ostry (2008) study fiscal reaction functions for a panel of multiple countries – similar results.
- Ghosh et al. (2013) show that ρ is much lower at high levels of debt – evidence of fiscal fatigue.
- But does the FRF remain stable over time? D'Erasmo et al. (2016) investigate.

MULTIPLE PATHS, SAME SOLVENCY

- Combine $b_t = (\bar{R}/\bar{\Gamma})\, b_{t-1} - s_t$ with the linear FRF $s_t = \mu_t + \rho\, b_{t-1} + \varepsilon_t$:

$$b_t = -\mu_t + \left(\frac{\bar{R}}{\bar{\Gamma}} - \rho\right) b_{t-1} - \varepsilon_t.$$

- Debt is stationary only if $\rho > \bar{R}/\bar{\Gamma} - 1$. But IGBC holds for any $\rho > 0$: debt grows slower than $\bar{R}/\bar{\Gamma}$, so the discounted value vanishes.
- Implication: halving ρ or lowering the intercept μ yields a *different* long-run debt ratio and different transitional dynamics, but the *same* present value of surpluses. The classic DSA cannot distinguish between these paths.

STRUCTURAL INSTABILITY OF THE FRF

- D'Erasmo, Mendoza, and Zhang (2016) re-estimate Bohn's FRF for the U.S. using data from 1791 to 2014.
- Pre-2008 sample: $\hat{\rho} \approx 0.10$, consistent with Bohn (2008).
- Full sample including 2008–2014: $\hat{\rho}$ drops to ≈ 0.08. The Chow test rejects parameter stability.
- FRF residuals for 2009–2011 are twice the minus-two-standard-error bound. The FRF predicted rising primary surpluses; the data showed persistent, large deficits.
- This pattern of prolonged primary deficits after a debt surge is unprecedented in U.S. fiscal history – after the Civil War, WWI, and WWII, surpluses appeared within three years.

FISCAL REACTION FUNCTIONS

- Leeper and Li (2017) warn against using surplus-debt regressions to assess debt sustainability.
- For the estimator of ρ to be consistent, we must have

$$\mathbb{E}(\epsilon_t \mid b_{t-1}) = 0.$$

  1. This means that shocks at $t-1$ that affect b_{t-1} must not affect ϵ_t.
  2. This means that b_{t-1} cannot depend on the expectation of ϵ_t.
- Since the value of debt depends on the expected value of future surpluses, this is a strong assumption: ϵ_t could be serially correlated.

TAKEAWAYS

- Classic DSA: whether debt is sustainable depends on $\bar{R}$ versus $\bar{\Gamma}$ and the primary balance – but the framework is silent on risk and cannot rank fiscal paths.
- The IGBC (forward-looking valuation) is the right framework, but checking it requires the correct stochastic discount factor, not the risk-free rate.
- Bohn's fiscal reaction function offers an elegant shortcut: if $\rho > 0$, the IGBC holds. But the FRF may be structurally unstable and the estimator may be biased.
- Next lecture: the asset-pricing approach to debt sustainability – how to measure the correct SDF and risk-adjusted fiscal space.